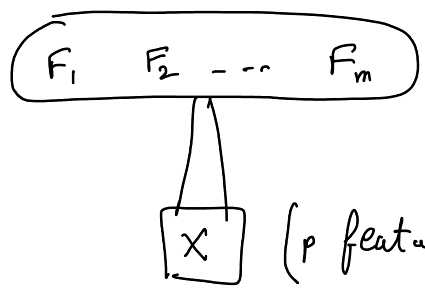
22 Oct

22 October 2025 10:13

Factor Model (orthognal factor model)



(p features)

$E(\underset{\sim}{F}) = 0$

$E(\underset{\sim}{e}) = 0$

$Cov(\underset{\sim}{F})_{m \times m} = I$

$Cov(\underset{\sim}{e})_{p \times p} = dig(\psi_i)$

F_j's are independent rv

e_i's are " "

F_j & e_i are independent

$$\underset{p\times 1}{X} = \underset{p\times 1}{\mu} + \underset{(p\times m)}{L}\underset{m\times 1}{F} + \underset{p\times 1}{e}$$

L: Loading matrix

e ← feature specific error vector / noise

Existence?

Uniqueness?

Find m?

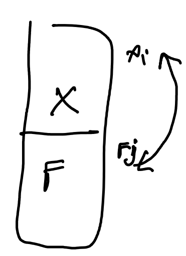
how to estimate L, e from sample data?

μ is not imp. because you can shift your data to μ.

Result for an m-factor model

$$\underset{\text{population}}{\Sigma} = \underset{p\times p}{Cov(X)} = LL^T + \psi$$

$$Cov(\underset{p\times 1}{\underset{\sim}{X}}, \underset{m\times 1}{\underset{\sim}{F}}) = \underset{p\times m}{L}$$



proof $X = \mu + LF + e$

$$Cov(X) = E\left((X-\mu)(X-\mu)^T\right)$$

$$= E\left((L\underset{\sim}{F} + e)(L\underset{\sim}{F} + e)^T\right)$$

$$= E\left((L\underset{\sim}{F} + \underset{\sim}{e})(F^T L^T + e^T)\right)$$

$$= E(L\underset{\sim}{F}\underset{\sim}{F}^T L^T) + E(\underset{\sim}{e}F^T L^T) + E(LFe^T) + E(ee^T)$$

$$= LE\left((F-0)(F-0)^T\right)L^T$$

$$+ E(e-0)(e-0)^T$$

F, e are indendet vector

$$E(F, e^T) = E((F-0)(e-0)^T)$$

$$= Cov(F, e)$$

$$= L I_m L^T + \psi = L L^T + \psi$$

Result for an m-factor model

$$\underset{p\times p}{\Sigma} = Cov(X) = LL^T + \psi$$

$$= Cov(X, F) = L \qquad = Cov(X, f)$$

$$= E\left[(X-\mu)(F-0)^T\right)$$

$$= E\left(\left(LF + e\right)F^T\right)$$

$$= LE(FF^T) + E(eF^T)$$

$$= LE(F-0)\,E(E-0)^T$$

$$= L\,Cov(\underset{\sim}{f}) + E\,Cov(e, F)$$

$$= L I_m + 0$$

$$= L$$

$(i,j)^{th}$ elem in $\Sigma = \underset{p\times m}{L}\underset{m\times p}{L^T} + \psi$

$$\sigma_{ii} = (e_{i1} \cdots e_{im})\begin{pmatrix} e_1 \\ \vdots \\ e_{im}\end{pmatrix} =$$

$$Cov(X_i, X_j) = \sigma_{ji} = \left(l_{i} \cdots l_{im}\right)_{1\times m}\begin{pmatrix} e_1 \\ \vdots \\ e_m \end{pmatrix} + 0$$

Communality of i^{th} var on m-factors

$$\sigma_{ij} = \sum_{k=1}^{m} l_{ik}\, l_{kj} \qquad i \neq j$$

Existense $\rightarrow$ depends on m

for a given m, there can be possibility that m-factor model does not exist.

$$\Sigma = \begin{bmatrix} 1 & 0.9 & 0.7 \\ 0.9 & 1 & 0.4 \\ 0.7 & 0.4 & 1 \end{bmatrix} \leftarrow \text{cor. matrix of sem } \cancel{\text{stu}}$$

$$\Sigma = \begin{pmatrix} l_{11} \\ l_{21} \\ l_{31} \end{pmatrix}\begin{pmatrix} l_{11} & l_{21} & l_{31}\end{pmatrix} + \begin{pmatrix} \psi_1 & 0 & 0 \\ 0 & \psi_2 & 0 \\ 0 & 0 & \psi_3 \end{pmatrix}$$

$$l_{22}^2 + \psi_2$$

$l_{11}^2 + \psi_1 = 1$ $\quad$ $l_{11}\, l_{31} = 0.7$

$l_{22}^2 + \psi_2 = 1$ $\quad$ $l_{21}\, l_{11} = 0.9$

$l_{33}^2 + \psi_3 = 1$ $\quad$ $l_{21}\, l_{31} = 0.4$

Parth se Notes mango and meko bhi bhej do pls

– subhojit